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RESEARCH:

Research on DeepFashion Datasets

Introduction

DeepFashion is one of the most important and widely used datasets in the field of fashion analysis and computer vision. It was developed to support artificial intelligence applications related to clothing recognition, garment detection, clothing segmentation, fashion recommendation systems, and virtual try-on applications. The dataset provides a large collection of fashion images along with detailed annotations, making it useful for training and evaluating AI models.

Purpose of DeepFashion Dataset

The main purpose of DeepFashion is to provide a standardized dataset for developing intelligent fashion systems. General image datasets usually focus on common objects such as cars, animals, and buildings, whereas DeepFashion specifically concentrates on clothing-related information.

By providing annotated fashion images, DeepFashion enables AI systems to understand garments and perform clothing analysis tasks.

Features of DeepFashion Dataset

DeepFashion contains several features that make it suitable for cloth segmentation research.

Large Collection of Fashion Images

DeepFashion contains more than 800,000 fashion images collected from online shopping websites and fashion platforms. The images include different clothing styles, viewpoints, poses, backgrounds, and lighting conditions.

This diversity improves model performance and helps AI systems learn various clothing patterns.

Rich Annotations

The dataset provides detailed annotations for each image. These annotations include:

- Clothing categories
- Bounding boxes
- Landmark information

- Clothing attributes
- Segmentation masks

These annotations help AI systems identify clothing boundaries and garment regions accurately.

Variants of DeepFashion Dataset

DeepFashion has evolved into different versions to support advanced fashion AI applications.

DeepFashion

The original DeepFashion dataset was mainly introduced for clothing recognition and retrieval tasks. It contains approximately 800,000 images and provides annotations related to clothing categories, attributes, and landmarks.

Although useful for recognition tasks, it provides limited segmentation information.

information.

DeepFashion-MultiModal Dataset

DeepFashion-MultiModal combines image data with textual information.

The dataset contains:

- Clothing images
- Product descriptions
- Fashion attributes
- Text metadata

This version supports multimodal AI applications such as recommendation systems and fashion search engines.

Dataset Structure and Annotations

DeepFashion datasets usually contain the following components.

Images

Fashion images collected from online stores and user photographs are provided. The images contain different clothing styles and human poses.

Bounding Boxes

Bounding boxes indicate garment locations within an image and help object detection systems identify clothing regions.

Landmarks

Landmarks represent important garment points such as collars, sleeves, waistlines, and hems. These improve garment alignment and pose estimation.

Segmentation Masks

Segmentation masks assign labels to clothing pixels and separate garments from body regions and backgrounds.

These masks are essential for clothing segmentation applications.

Applications of DeepFashion Dataset in Cloth Segmentation AI

DeepFashion datasets are widely used in fashion AI systems.

Clothing Extraction

AI systems trained on DeepFashion can separate clothing from human body regions.

Upper and Lower Wear Segmentation

The dataset supports identification of upper garments such as shirts and jackets and lower garments such as pants and skirts.

Virtual Try-On Systems

DeepFashion helps create digital dressing systems where users can visualize clothes before purchasing.

Fashion Recommendation Systems

AI models analyze clothing styles and recommend similar products.

Smart Retail Systems

Retail platforms use DeepFashion for product classification, garment tagging, and inventory management.

Conclusion

DeepFashion datasets play a significant role in fashion-based artificial intelligence and clothing segmentation systems. They provide large-scale annotated fashion images that help AI models understand clothing structures, garment categories, and segmentation boundaries.

Research on Segmentation Models

Introduction

Segmentation models are deep learning models used in computer vision to divide an image into meaningful regions by assigning labels to individual pixels. Unlike image classification, which predicts a single label for the entire image, segmentation identifies object boundaries and separates different regions within an image.

Need for Segmentation Models in Cloth Segmentation AI

Traditional object detection methods usually identify clothing items using bounding boxes. However, they cannot accurately separate garments from body regions or backgrounds.

Segmentation models overcome this limitation by classifying every pixel in an image. This allows AI systems to determine which pixels belong to clothes, body parts, and background regions.

Types of Image Segmentation

Segmentation techniques are generally classified into different categories.

Semantic Segmentation

Semantic segmentation classifies every pixel according to predefined classes.

All pixels belonging to the same category receive the same label.

Semantic segmentation is useful for:

- Clothing detection
- Upper and lower wear segmentation
- Garment masking

Common semantic segmentation models include:

- U-Net
- Fully Convolutional Network (FCN)
- DeepLab

Instance Segmentation

Instance segmentation identifies individual objects separately even if they belong to the same category.

For example, if an image contains two jackets, semantic segmentation labels both as "jacket," while instance segmentation creates separate masks for each jacket.

Panoptic Segmentation

Panoptic segmentation combines semantic and instance segmentation.

This method provides detailed image understanding and supports advanced fashion AI applications.

Popular Segmentation Models Used in Cloth Segmentation AI

Fully Convolutional Network (FCN)

Fully Convolutional Network (FCN) was one of the earliest deep learning models developed for semantic segmentation.

Unlike traditional convolutional neural networks, FCN removes fully connected layers and performs pixel-wise prediction using convolution operations.

FCN provides pixel-level outputs but has limitations in preserving fine boundaries.

U-Net

U-Net is one of the most popular segmentation architectures used in image analysis.

It consists of two parts:

Encoder

The encoder extracts image features using convolution and pooling operations.

Decoder

The decoder reconstructs image information and produces segmentation masks.

U-Net uses skip connections that transfer feature information from encoder layers to decoder layers, improving segmentation accuracy.

Mask R-CNN

Mask R-CNN extends Faster R-CNN by adding a segmentation branch.

The model performs three operations simultaneously:

1. Object detection
2. Classification
3. Mask generation

Mask R-CNN is useful in fashion applications because it separates individual garments independently.

DeepLab Models

DeepLab is a family of semantic segmentation models developed for accurate image understanding.

DeepLab introduces important techniques such as:

Atrous Convolution

Atrous convolution increases the receptive field without increasing computational complexity.

This helps capture larger contextual information.

Spatial Pyramid Pooling

Spatial pyramid pooling extracts information at multiple image scales.

This improves segmentation quality for garments of different sizes.

DeepLab V3+ is widely used in clothing segmentation because it provides accurate garment boundaries.

SegFormer

SegFormer is a transformer-based segmentation model.

Unlike CNN models, SegFormer uses transformer mechanisms to understand image relationships.

Workflow of Segmentation Models

The general workflow of segmentation models includes multiple stages.

Image Input

Fashion images containing people wearing clothes are provided to the model.

Mask Generation

The model generates segmentation masks to isolate garment regions.

Output Production

The final segmented image is produced.

Conclusion

Segmentation models are fundamental components of Cloth Segmentation AI because they enable pixel-level understanding of garments. Models such as FCN, U-Net, Mask R-CNN, DeepLab, and SegFormer provide different approaches for clothing detection and segmentation.

Tasks:

Clothing Segmentation Tasks in Cloth Segmentation AI

Separate Clothing from Body

Introduction

Separating clothing from the human body is one of the fundamental tasks in Cloth Segmentation AI. The objective is to identify clothing regions and distinguish them from body parts such as skin, face, hands, and other visible human regions.

This task is important because many fashion applications require garments to be extracted independently from the person wearing them.

Examples include:

- Virtual try-on systems
- Clothing recommendation systems
- Garment analysis tools
- Image editing applications

Clothing-body separation enables AI systems to isolate garments and improve further processing.

Working Process

The separation process generally follows multiple stages.

Image Input

The system receives an image containing a person wearing clothes.

The image may contain:

- Upper garments
- Lower garments
- Skin regions
- Background objects

Feature Extraction

The model extracts clothing features such as:

- Colors
- Textures
- Edges
- Garment boundaries

Pixel Classification

The segmentation model labels pixels as:

- Clothing
- Body
- Background

Mask Generation

Clothing masks are generated while removing:

- Face
- Skin
- Hair
- Background

This produces isolated clothing regions.

Detect Garment Areas

Introduction

Garment area detection focuses on identifying and locating clothing items present within an image. The objective is to determine the position and boundaries of garments such as shirts, jackets, dresses, and pants.

This task acts as an intermediate step before segmentation because garment locations are detected before detailed mask generation.

Garment area detection supports:

- Clothing analysis
- Fashion recommendation
- Product identification
- Clothing extraction

Working Process

Image Acquisition

An input image containing clothing items is provided.

Feature Extraction

The model extracts information including:

- Texture

- Color
- Clothing shape
- Edges

Garment Identification

The AI model identifies clothing categories.

Example:

- Shirt detected
- Pants detected
- Jacket detected

Region Detection

The system identifies garment positions using:

- Bounding boxes
- Segmentation masks
- Pixel regions

This produces garment location outputs.

Types of Garments Detected

Upper Garments

Examples:

- Shirt
- T-shirt
- Jacket
- Sweater

Lower Garments

Examples:

- Pants
- Jeans
- Shorts
- Skirts

Full Body Garments

Examples:

- Dresses
- Gowns
- Overalls

Segment Upper and Lower Wear

Introduction

Upper and lower wear segmentation focuses on separating garments according to body position. The objective is to distinguish upper-body garments from lower-body garments and generate separate segmented outputs.

This process improves clothing understanding and enables category-based garment analysis.

Upper Wear Segmentation

Upper wear segmentation identifies garments worn on the upper body.

Examples include:

- Shirt
- Jacket
- Coat
- Hoodie
- Sweater

The segmentation model identifies upper garment pixels and generates masks.

Example output:

Upper wear → Shirt mask generated

Lower Wear Segmentation

Lower wear segmentation identifies garments worn on the lower body.

Examples include:

- Jeans
- Pants
- Shorts
- Skirts

- Trousers

The model generates separate masks for lower garments.

Example output:

Lower wear → Jeans mask generated

Workflow

Image Input

Fashion image containing clothing items is provided.

Feature Extraction

The model extracts:

- Texture
- Color
- Shape
- Garment boundaries

Pixel Classification

Labels assigned:

Shirt → Upper wear

Jacket → Upper wear

Jeans → Lower wear

Skirt → Lower wear

Background → Background

Mask Generation

Separate outputs are generated:

Upper mask → Upper garments

Lower mask → Lower garments

Applications

Upper-lower wear segmentation is used in:

- Virtual try-on systems
- Fashion recommendation platforms

- Smart retail applications
- Clothing analysis systems
- Image editing tools

Importance of Clothing Segmentation Tasks

The combination of clothing-body separation, garment detection, and upper-lower wear segmentation improves the overall performance of Cloth Segmentation AI systems.

These tasks help AI models:

- Understand clothing structure
- Extract garments accurately
- Generate segmentation masks
- Support fashion applications

Together, these tasks form the foundation of clothing segmentation systems.

Conclusion

Clothing segmentation tasks are essential components of Cloth Segmentation AI. Separating clothing from the body enables garment extraction, garment area detection identifies clothing regions, and upper-lower wear segmentation performs category-based separation.

These tasks improve clothing understanding and support applications such as virtual try-on systems, recommendation engines, smart retail systems, and garment analysis platforms. Accurate clothing segmentation forms the basis for future fashion AI technologies.

Clothing Segmentation Prototype Using OpenCV and MediaPipe

Introduction

A real-time clothing segmentation prototype was developed using **OpenCV** and **MediaPipe Selfie Segmentation**. The prototype uses webcam input to capture live video frames and performs segmentation to separate the human region from the background.

The system demonstrates image masking and segmentation concepts by generating masks in real time and displaying segmented outputs. The prototype helps understand clothing extraction workflows and serves as a basic implementation of Cloth Segmentation AI.

Objective of the Prototype

The prototype was developed with the following objectives:

- Capture live webcam video
- Perform real-time segmentation
- Generate segmentation masks
- Apply image masking
- Separate foreground from background
- Demonstrate clothing extraction workflow

Tools and Libraries Used

OpenCV (cv2)

OpenCV was used for:

- Webcam access
- Video frame processing
- Displaying output windows
- Image manipulation

MediaPipe Selfie Segmentation

MediaPipe Selfie Segmentation was used to generate segmentation masks.

Function used:

mp.solutions.selfie_segmentation

The model identifies foreground regions and produces pixel-level masks.

NumPy

NumPy was used for masking operations and background generation.

Prototype Workflow

The developed system follows the workflow below:

Webcam Input → Frame Processing → Segmentation → Mask Generation → Background Removal → Output Display

Working Procedure

Step 1: Webcam Initialization

The webcam was initialized using:

cap = cv2.VideoCapture(0)

The camera continuously captured live frames.

Step 2: Frame Preprocessing

Captured frames were:

- Flipped horizontally

frame = cv2.flip(frame,1)

- Converted from BGR to RGB format

rgb = cv2.cvtColor(
frame,
cv2.COLOR_BGR2RGB
)

RGB format was required because MediaPipe processes RGB images.

Step 3: Segmentation Processing

MediaPipe segmentation model processed the frame:

result = segmenter.process(rgb)

The output produced:

segmentation.py > ...

```python
import cv2
import mediapipe as mp
import numpy as np
mp_selfie = mp.solutions.selfie_segmentation
segmenter = mp_selfie.SelfieSegmentation(
    model_selection=1
)
cap = cv2.VideoCapture(0)
while cap.isOpened():
    ret, frame = cap.read()
    if not ret:
        break
    frame = cv2.flip(frame, 1)
    rgb = cv2.cvtColor(
        frame,
        cv2.COLOR_BGR2RGB
    )
    result = segmenter.process(rgb)
    mask = result.segmentation_mask
    condition = mask > 0.5
    bg = np.ones_like(frame) * 255
    segmented = np.where(
        condition[:, :, None],
        frame,
        bg
    )
    cv2.imshow(
        "Original Webcam",
        frame
    )
    cv2.imshow(
        "Segmentation Mask",
        mask
    )
    cv2.imshow(
        "Clothing Segmentation Prototype",
        segmented
    )
    if cv2.waitKey(1) & 0xFF == ord('q'):
        break
cap.release()
cv2.destroyAllWindows()
```

mask = result.segmentation_mask

The mask contained pixel probabilities for foreground regions.

Step 4: Image Masking

Thresholding was applied:

condition = mask > 0.5

A white background was created:

bg = np.ones_like(frame)*255

This removed background regions while preserving foreground objects.

Output Windows Generated

The prototype generated three windows.

Original Webcam

Displays live webcam feed.

Segmentation Mask

Displays generated segmentation masks.

White region → Foreground

Dark region → Background

Clothing Segmentation Prototype

Displays segmented output after masking.

Background is removed while foreground remains visible.

Prototype Output

Input:

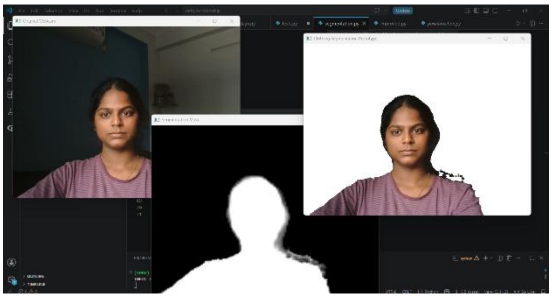
Live webcam feed showing a person wearing clothes.

Output:

- Segmentation mask generated
- Background removed

- Human region isolated
- Clothing visible in segmented frame

The system successfully demonstrated image masking and segmentation.



Observations

1. Real-time segmentation was achieved successfully.
2. MediaPipe generated segmentation masks quickly.
3. Background removal improved clothing visibility.
4. The prototype worked efficiently using webcam input.
5. Clothing regions remained visible after masking.

Limitations

The prototype has certain limitations:

- Individual garments are not segmented separately.
- Upper and lower wear classification is not performed.
- Similar foreground and background colors may affect results.
- Complex poses reduce segmentation quality.

Conclusion

The clothing segmentation prototype successfully demonstrated real-time image masking and segmentation using OpenCV and MediaPipe. The webcam-based implementation captured live frames, generated segmentation masks, and removed background regions.

The prototype validated the basic workflow of Cloth Segmentation AI and provided a foundation for future developments involving garment detection and clothing segmentation.